

Labor demand. With perfectly competitive labor markets, workers are paid their marginal revenue product:

$$W_j = \frac{\partial P_j Y_j}{\partial H_j} = (1 - 1/\epsilon) Y_j^{-\frac{1}{\epsilon}}. \quad (16)$$

Since firms' production and demand curves are symmetric and workers derive the same net benefit $\hat{\alpha}_1$ from encouragement across employers, two skill prices exist in equilibrium: W_1 at encouraged firms and W_0 at non-encouraged firms. Let $j \in \{0, 1\}$ indicate whether a firm encourages chatbot use.

Employer encouragement. Employers derive log utility from flow profits and encourage chatbot use if the resulting profit gain exceeds their encouragement cost κ_j :

$$E_j = \mathbf{1} [\kappa_j \leq \log \Pi_j(1) - \log \Pi_j(0)]. \quad (17)$$

Constant returns to scale and CES demand imply that optimal flow profit is $\Pi_j = \frac{1}{\epsilon} Y_j^{1-1/\epsilon}$, so the profit gain from encouragement equals:

$$\log \Pi_j(1) - \log \Pi_j(0) = (1 - \epsilon)(\log W_1 - \log W_0). \quad (18)$$

Since this gain is common across employers, encouragement follows a cutoff rule:

$$E_j = \mathbf{1} [\kappa_j \leq \kappa], \quad \text{with} \quad \kappa = (1 - \epsilon)(\log W_1 - \log W_0). \quad (19)$$

Adoption. Workers derive log utility from earnings and adopt chatbots if the monetary benefit exceeds the adoption cost. Fixing the employer, the adoption rule reduces to:

$$A_{ij} = \mathbf{1} [B_{ij} \geq C_{ij}] = \mathbf{1} [\hat{\alpha}_0 + \hat{\alpha}_1 E_j + \hat{\beta} \times i \geq 0], \quad (20)$$

where $\hat{x} = x_b - x_c$ denotes net benefits, yielding the cutoff:

$$A_{ij} = \mathbf{1} [i \leq i^*(E_j)], \quad \text{with} \quad i^*(E) = -\frac{1}{\hat{\beta}}(\hat{\alpha}_0 + \hat{\alpha}_1 E). \quad (21)$$

Sorting. Workers also choose which firms to sort into. Since all adopters receive the same net benefit $\hat{\alpha}_1$ from encouragement, workers effectively choose between sorting into encouraged firms and adopting, or sorting into non-encouraged firms and not adopting:

$$E_{j(i)} = \mathbf{1} \left[\hat{\alpha}_0 + \hat{\alpha}_1 + \hat{\beta} \times i + \log W_1 - \log W_0 \geq 0 \right], \quad (22)$$

implying the cutoff:

$$E_{j(i)} = \mathbf{1} [i \leq i^*], \quad \text{with} \quad i^* = -\frac{1}{\hat{\beta}}(\hat{\alpha}_0 + \hat{\alpha}_1 + \log W_1 - \log W_0). \quad (23)$$

A.2.3 Equilibrium

An equilibrium consists of encouragement policies $\{E_j\}$, skill prices $\{W_j\}$, worker-firm allocations $\{\mathcal{I}_j\}$, and adoption decisions $\{A_{ij}\}$, such that firms and workers optimize (Equations (16), (17), (21), and (23)) and markets clear (Equations (14) and (15)). Without chatbots, all firms are symmetric and each employs an equal share of baseline human capital.

Sorting. When chatbots are available, the equilibrium features full segregation: assuming $\hat{\alpha}_1 \geq 0$, all adopters sort into encouraged firms and all non-adopters into non-encouraged firms, or both. We focus on the empirically relevant case where some but not all firms encourage and some but not all workers adopt.

Skill prices. From Equation (19), equilibrium skill price differentials satisfy:

$$\log W_1 - \log W_0 = \frac{\kappa}{1 - \epsilon}, \quad (24)$$

where κ is the encouragement cost of the marginal firm. The cost of encouragement is thus priced into skill differentials: workers are not rewarded for these firm-level investments.

For analytical simplicity, we focus on the case where all employers face the same encouragement cost, $\kappa_j = \kappa$ for all j . All firms are then marginal, and the share of encouraged employers adjusts to match the relative supply of adopting workers' human

capital.²⁵

A.2.4 Model Predictions

We now derive predictions for how chatbot arrival affects labor market outcomes.

Worker earnings. Combining Equations (10)–(11), the log change in earnings of worker i at firm j is:

$$d \log Y_{ij} = \underbrace{d \log W_j}_{\text{skill-price effect}} + \underbrace{A_i B_{ij}}_{\text{productivity effect}} \quad (25)$$

$$= d \log W_0 + E_j \frac{\kappa}{1 - \epsilon} + A_i(\alpha_{b0} + \beta_b \times i) + E_j A_i \alpha_{b1}. \quad (26)$$

This yields three key predictions:

1. Non-adopters experience an earnings loss at encouraged firms:

$$\log Y_{i1}(A_i = 0) - \log Y_{i0}(A_i = 0) = -\frac{\kappa}{\epsilon - 1} \leq 0 \quad (27)$$

2. Holding encouragement fixed, adoption increases earnings by the productivity effect:

$$\log Y_{iE}(A_i = 1) - \log Y_{iE}(A_i = 0) = \alpha_{b0} + \alpha_{b1} \times E + \beta_b \times i \quad (28)$$

3. The effect of encouragement on adopter earnings is ambiguous, depending on the relative magnitudes of the productivity and skill price effects:

$$\log Y_{i1}(A_i = 1) - \log Y_{i0}(A_i = 1) = \alpha_{b1} - \frac{\kappa}{\epsilon - 1} \gtrless 0. \quad (29)$$

Even if encouragement enhances productivity ($\alpha_{b1} > 0$, as documented in Section 2.2), these gains need not translate into higher earnings when encouragement is costly (high κ), demand is inelastic (low ϵ), or productivity gains are modest (low α_{b1}).

²⁵With heterogeneous encouragement costs, skill price differentials also depend on the distribution of κ_j through the supply and demand for human capital across encouraged and non-encouraged firms. The key qualitative predictions are unchanged: encouragement is priced into skill differentials, and the sign of the earnings effect for encouraged adopters remains ambiguous.

Adopters may still prefer encouraged firms if encouragement sufficiently reduces adoption costs (i.e., $\alpha_{c1} < \alpha_{b1} - \frac{\kappa}{\epsilon-1}$), but these benefits will not appear in earnings.

Occupational mobility. If chatbots create greater benefits B_{ij} for newcomers to an occupation, adoption will correlate with occupational switching (as documented in Section 3.3).²⁶

Worker-firm sorting. Following Equation (23), chatbots trigger reallocation of workers who benefit from adoption (low- i types) into encouraged firms ($E = 1$).

Worker outcomes. Figure A.2 summarizes how chatbots affect sorting, adoption, and earnings. The gray line traces equilibrium patterns: workers with $i < i^*$ sort into encouraged firms and adopt, while those with $i > i^*$ sort into non-encouraged firms and do not adopt. The figure illustrates the case where encouragement increases adopter earnings; as shown above, encouragement can also reduce them, and if the amenity value of encouraged use is large enough (α_{c1} sufficiently low), some encouraged adopters may even earn less than non-encouraged non-adopters.

²⁶This prediction reflects an effect running from occupational choice to adoption. Causality may also run the other way, with chatbots prompting moves into new occupations; our framework abstracts from this margin.